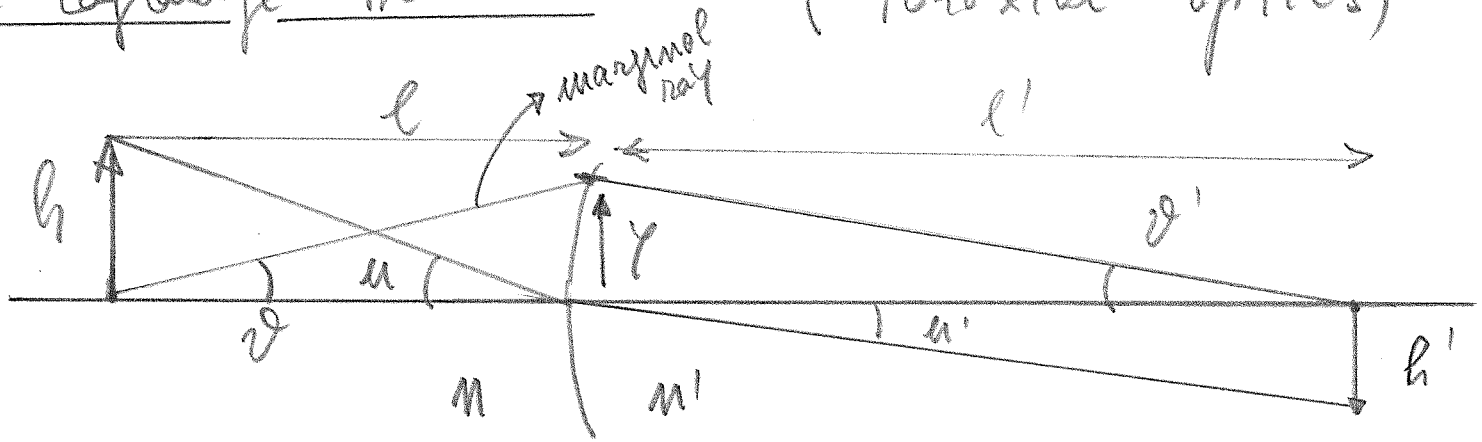
The Lagrange theorem (paraxial optics)



$$n \sin(u) = n' \sin(u') \rightarrow n u = n' u'$$

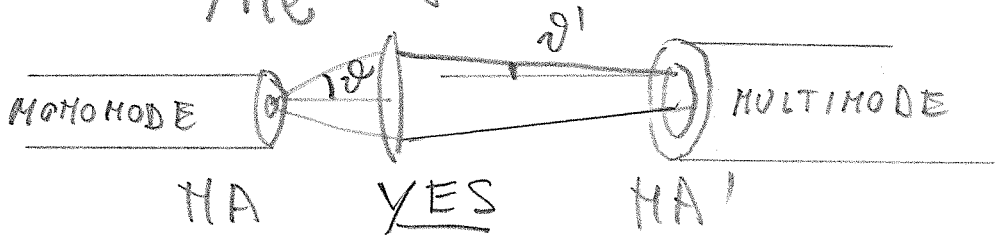
$$n h/e = n' h'/e' \rightarrow n h (Y/e) = n' h' (Y/e')$$

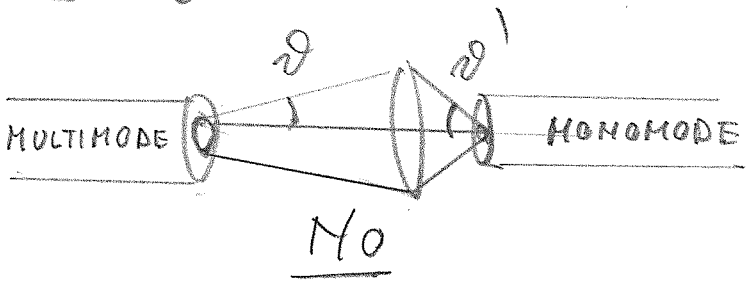
$$\vartheta \qquad \vartheta'$$

$$\underbrace{n h \vartheta}_{\text{optical invariant}} = n' h' \vartheta' \rightarrow \text{Lagrange theorem}$$

$$M_e = h'/h \qquad M_\vartheta = \vartheta'/\vartheta \qquad M_e \cdot M_\vartheta = n'/n$$

$$M_\vartheta = 1/M_e \text{ if } n = n' \rightarrow h' \vartheta' = h \vartheta$$





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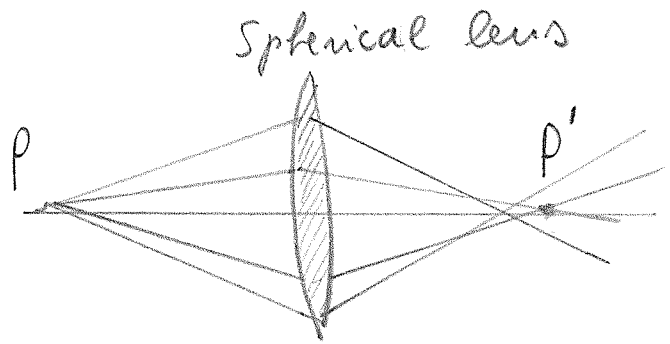
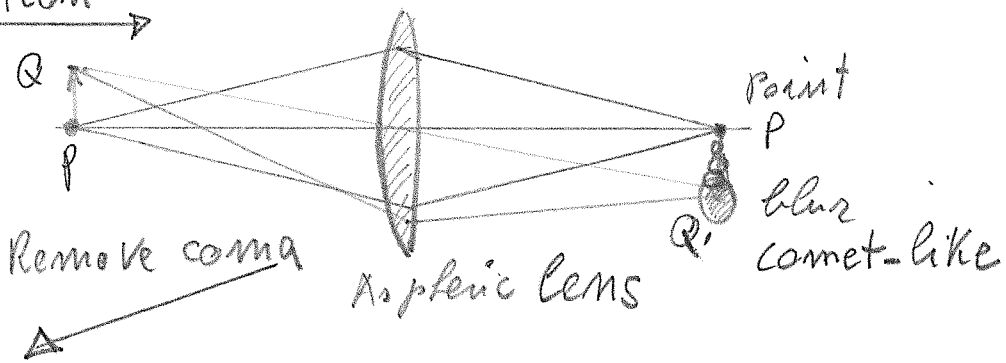
Abbe's sine condition

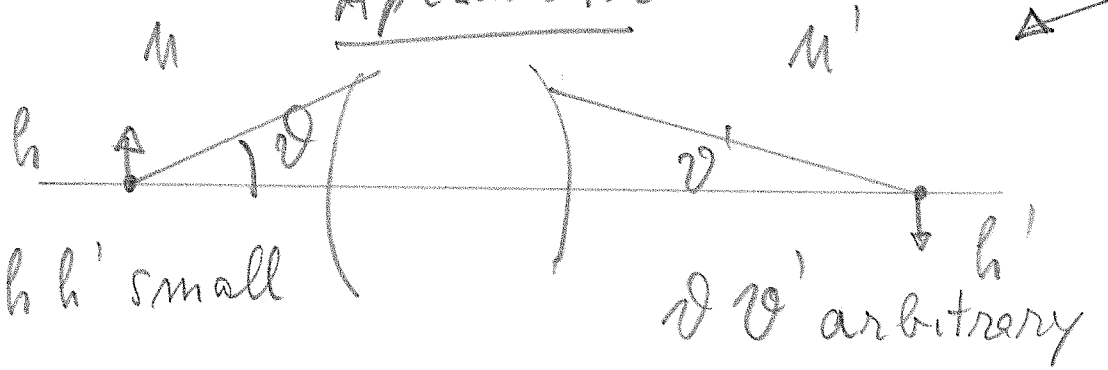


Remove spherical aberration →

Stigmatic



Aplanatic



$$n h \sin\vartheta = n' h' \sin\vartheta'$$

$$(n h \vartheta = n' h' \vartheta')$$

Aplanatic systems are free from spherical aberration and coma

Coma depends on the different magnification factors of the annular regions of the lens at different distance from the centre

Aberration free optical systems
- → Paraxial approx. → Optical invariant $[n h \vartheta = n' h' \vartheta']$
- → Wide aperture → Abbe's condition $[n h \sin\vartheta = n' h' \sin\vartheta']$

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The radiometric quantities

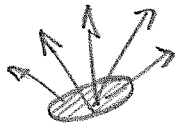
Assumption:

1) geometrical optics → no aberrations, no diffraction
2) No spectral discrimination → total power of the radiation $\forall \lambda$
3) Propagation in lossless media of incoherent beams

Quantities

→ Radiant flux → Total power of the beam ϕ [W] emitted by a source and received by a det.

→ Radiant exitance → density of radiant flux emitted by an extended source per unit area at any angle $M(x,y) = \frac{d\phi}{dA}$ [W/m²]



→ Irradiance → density of radiant flux received by a real or virtual surface per unit area from any angle $E(x,y) = \frac{d\phi}{dA}$ [W/m²]

- Radiant intensity → density of radiant flux emitted by a small source per unit solid angle in a given direction $I(\vartheta,\varphi) = \frac{d\phi}{d\Omega}$ [W/sr]

- Radiance → density of radiant flux emitted or received per unit solid angle and unit of projected area along the direction of observation $L(x,y;\vartheta,\varphi) = \frac{d\phi}{dA_\perp d\Omega}$ [W/m² sr]

The radiance is a property of a radiation field an is conserved in lossless media

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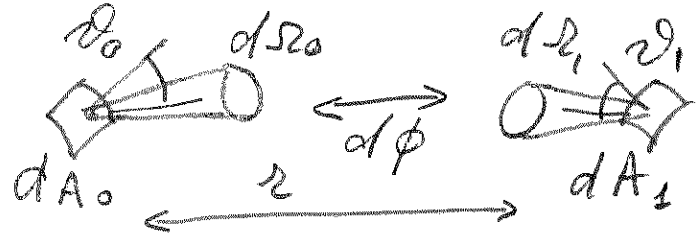
Law of conservation of the radiance

Free space propagation (homogeneous lossless medium) → No absorption & scattering



Etendue of a radiation field (source)

$$d\mathcal{E}_0 = dA_0 \cos\vartheta_0 \, d\Omega_0 \quad \text{(invariant)}$$

$$L_0 = \frac{d\phi}{d\mathcal{E}_0} \qquad d\phi_0 = L_0 \, dA_0 \cos\vartheta_0 \, d\Omega_0$$

$$d\Omega_0 = \frac{dA_1 \cos\vartheta_1}{r^2}$$

$$d\Omega_1 = \frac{dA_0 \cos\vartheta_0}{r^2}$$

$$L_1 = \frac{d\phi}{dA_1 \cos\vartheta_1 \, d\Omega_1} = \frac{L_0 \, \cancel{dA_0 \cos\vartheta_0} \, d\Omega_0}{dA_1 \cos\vartheta_1 \, \cancel{dA_0 \cos\vartheta_0}} \cdot r^2 = \frac{L_0 \, \cancel{dA_1 \cos\vartheta_1}}{\cancel{r^2}} \, \frac{\cancel{r^2}}{\cancel{dA_1 \cos\vartheta_1}} = L_0$$

$$L_1 = L_0$$

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